

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi

CS406: Mid-semester Examination
25th January 2023

Max. marks: 20

Time: 1 hr

Q 1. Find the real root of the equation $3x = \cos x + 1$, using Newton-Raphson method.

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Q 2. Solve the following system of equations using the Gauss-Jordan method

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$$\begin{aligned}x + y + z &= 1 \\4x + 3y - z &= 6 \\3x + 5y + 3z &= 4\end{aligned}$$

Q 3. Find the smallest positive root of $x^2 - 5x + 1 = 0$, correct to four decimal places

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Q 4. Test the consistency of the following system of equations

$$\begin{aligned}x + 10y - z &= 3 \\2x + 3y + 20z &= 7 \\9x + 22y + 79z &= 45.\end{aligned}$$

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Q 5. Using Lagrange interpolation, find the unique polynomial $P(x)$ of degree 2 or less such that $P(1) = 1$, $P(3) = 27$, $P(4) = 64$.

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End-semester Examination
Numerical Methods (CS-106)
7th March 2023

Time: 3 hours

Total Marks: 50

PART - A

Answer all questions. ($5 \times 4 = 20$ marks)

Q1. How many positive roots does the equation $3x^3 - 2x^2 - x - 5 = 0$ have?

- | | |
|-------|-------|
| (a) 2 | (b) 3 |
| (c) 1 | (d) 0 |

Q2. The positive real roots of the equation $x^3 - 3x + 1 = 0$ lie in the intervals:

- | | |
|---------------------|---------------------|
| (a) (0, 1) & (1, 2) | (b) (2, 1) & (1, 3) |
| (c) (0, 2) & (1, 3) | (d) (0, 1) & (2, 3) |

Q3. Using Newton Raphson's method, the the smallest positive root of the equation $f(x) = x^3 - 5x + 1 = 0$ is:

- | | |
|--------------|--------------|
| (a) 0.201640 | (b) 0.120768 |
| (c) 0.301640 | (d) 0.102640 |

Q4. The solution of the system of equations

$$x_1 + 10x_2 - x_3 = 3$$

$$2x_1 + 3x_2 + 20x_3 = 7$$

$$10x_1 - x_2 + 2x_3 = 4$$

using the Gauss elimination with partial pivoting.

- (a) $x_1 = 0.53752, x_2 = 0.48940, x_3 = 0.36918$
 (b) $x_1 = 0.37512, x_2 = 0.28940, x_3 = 0.26908$
 (c) $x_1 = 0.17512, x_2 = 0.38194, x_3 = 0.32690$
 (d) $x_1 = 0.27432, x_2 = 0.18953, x_3 = 0.13469$

Q5. Using Lagrange's formula, the quadratic polynomial that takes the following values are:

$$x_0 = 0, y_0 = 0$$

$$x_1 = 1, y_1 = 1$$

$$x_2 = 3, y_2 = 0$$

(a) $6x^2 + 4x - 1$

(b) $4x^2 - 3x + 2$

(c) $\frac{-1}{2}(x^2 - 3x)$

(d) $\frac{1}{2}(3x^2 - x)$

Part-B

Answer all questions. (10 marks each, 10 = 30)

Q6. The following data gives the velocity of a particle for 20 seconds at an interval of 5 seconds. Find the initial acceleration using the entire data.

Time (sec)	0	5	10	15	20
Velocity (m/sec)	0	3	14	69	228

Q7. Find an approximation to $y(1.6)$, for the initial value problem

$$y' = x + y^2, y(1) = 1$$

using the Euler method with $h = 0.1$.

Q8. Derive the trapezium rule using the Lagrange linear interpolating polynomial. Find the approximate value of

$$I = \int_0^1 f(x) dx$$

where

$$f(x) = \frac{1}{1+x}$$

using the trapezium rule with 4 equal subintervals.

$$l_0(x) = \frac{x-b}{a-b}$$

$$l_1(x) = \frac{x-a}{b-a}$$

$$I = \int_a^b f(x) dx$$